

Sri Lanka Institute of Information Technology



IT2130 – Operating Systems and System Administration Year
2, Semester 2- 2026

Practical Answer Submission

Practical Sheet No: 08

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Activity 01

- `#include <stdio.h>`

Includes standard input/output functions like `printf()`

- `#include <sys/ipc.h>`
`#include <sys/shm.h>`

Libraries for inter-process communication (IPC) and shared memory

- `#include <string.h>`

Used for string operations

- `int main() {`

Program execution starts here

- `key_t key = 1234;`

Unique key to identify shared memory segment

- `int shmid;`

Variable to store shared memory ID

- `char *ptr;`

Pointer to access shared memory

- `shmid = shmget(key, 4096, 0666 | IPC_CREAT);`

Creates shared memory segment;

4096 = size (bytes) 0666 =

read/write permission

IPC_CREAT = create if not exists

- `ptr = (char *) shmat(shmid, NULL, 0);`

Attaches shared memory to process and returns pointer

- `sprintf(ptr, "Hello from shared memory!");`

Writes message into shared memory

- `printf("Data written to shared memory\n");`

Displays confirmation message

- `shmdt(ptr);`

Detaches shared memory from process

- `return 0;`

}

Ends program

Activity 02

- `key_t key = 1234;`

Unique key to identify shared memory segment

- `shmid = shmget(key, 4096, 0666);`

Access existing shared memory (no IPC_CREAT)

- `ptr = (char *) shmat(shmid, NULL, 0);`

Attach memory

- `printf("Message from shared memory: %s\n", ptr);`

Read and display message

- `shmdt(ptr);`

Detach memory

- `shmctl(shmid, IPC_RMID, NULL);`

Deletes shared memory

Activity 03

```
GNU nano 7.2 lab8.c
#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <string.h>

int main() {

    key_t key = 1234;
    int shmid;
    char *ptr;

    shmid = shmget(key, 4096, 0666 | IPC_CREAT);

    ptr = (char *) shmat(shmid, NULL, 0);

    sprintf(ptr, "Operating Systems Lab");
    printf("Message written to shared memory\n");

    printf("Message read from shared memory: %s\n", ptr);

    shmdt(ptr);

    shmctl(shmid, IPC_RMID, NULL);

    return 0;
}
```

```
vboxuser@ubuntu:~$ nano lab8.c
vboxuser@ubuntu:~$ gcc -o lab8 lab8.c
vboxuser@ubuntu:~$ ./lab8
Message written to shared memory
Message read from shared memory: Operating Systems Lab
vboxuser@ubuntu:~$
```